Systematic Pair-Level Campaign for δ_{pair}: Convergence, Inter-Pair Concentration, and Normalization Structure

O25 — Spectral Admissibility Sub-programme

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Abstract

The O-series establishes the transfer chain $c_\chi \to \delta_{\mathrm{pair}} \to \beta^*$ unconditionally with respect to the fibre structure of the non-injective projection Π [13]. The value $\delta_{\mathrm{pair}} \approx 7.44$ extracted in O16 [4] was based on a single conjugate pair per prime and a limited prime range. The present paper reports the first systematic campaign computing δ_{pair} across *all* $(q-1)/2$ conjugate pairs $(c, q-c)$ for $q \in \{29, 61, 101, 151\}$, with $M = 50$ independent block samples per pair. Three results are established. First, $\delta_{\mathrm{pair}}(q)$ converges robustly: the inter-pair standard deviation decreases from 0.54 at $q = 29$ to 0.14 at $q = 151$, confirming that δ_{pair} is a structural invariant of the Weil representation rather than a block-level fluctuation. Second, empirical fits of the form $\delta_\infty + a/\log q$ describe the data excellently over the tested range, but carry no asymptotic meaning: the leading finite-size correction is governed by the BFS window depth $n_1(q)$, which satisfies $n_1(q) = \Theta(q)$ by the Window-Depth Theorem [8], making the correction approximately constant in $\log q$ and rendering direct extrapolation structurally degenerate. Third, when the O14 normalization correction $\eta \log q/\log n_1(q)$ [7] is applied, the corrected values $\delta_{\mathrm{corr}}(q) \in [7.7, 9.0]$ fall within the admissible window $[7.4, 10.6]$ consistently across $q \in \{61, 101, 151\}$. This supports the interpretation that the apparent decrease of raw $\delta_{\mathrm{pair}}(q)$ is a finite-size normalization effect rather than a signal of asymptotic drift below the admissible window. The identification of the asymptotic value δ_∞ requires analytical control of $n_1(q)/q$, identified here as the central open direction.

Keywords. Cosmochrony; spectral admissibility; pair-level observable; Weil representation; Heisenberg graphs; capacity exponent; convergence; inter-pair concentration; normalization correction; window depth; BFS; asymptotic analysis

Contents

1 Position of the Problem

1.1 What O16–O24 established

The spectral admissibility sub-programme has established, through O16–O24 [4, 5, 6, 1, 9, 11, 10, 12, 13], that the physically relevant observable is the canonical pair-level quantity

$$\sigma^{\mathrm{can}}_{\mathrm{pair}}(n) = \sigma_c(n)\,\sigma_{q-c}(n), \tag{1}$$

with capacity exponent $\delta_{\mathrm{pair}} \approx 7.44$ consistent with the phenomenological target $\delta_{\mathrm{pair}} \in [7.4, 10.6]$. The transfer chain $c_\chi \to \delta_{\mathrm{pair}} \to \beta^*$ was established unconditionally with respect to the fibre structure of Π in O24 [13].

The remaining open direction identified in O24 Section 5.1 is the extended numerical campaign at large primes across multiple conjugate pairs. O16 computed δ_{pair} for a single conjugate pair per prime (those pairs that happened to appear simultaneously in the O12 random block sample [3]). No systematic coverage of all pairs was available.

1.2 Scope of O25

The present paper provides the first systematic computation of δ_{pair} across all $(q-1)/2$ conjugate pairs for each prime, with $M = 50$ independent block samples per pair. This yields:

- the full distribution of δ_{pair} across pairs at each prime,
- the mean $\bar{\delta}_{\mathrm{pair}}(q)$ and inter-pair standard deviation σ_q,
- the q-dependence of these quantities over $q \in \{29, 61, 101, 151\}$.

The paper does *not* claim to identify the asymptotic value δ_∞. Instead, it establishes why direct extrapolation in q is structurally unreliable and identifies the correct analytical target: the ratio $n_1(q)/q$.

2 Setup and Protocol

2.1 Observable and block parameterisation

We follow the O12/O13 protocol [3, 2] without modification. For a prime q and a generic block $c_{\mathrm{block}} = (c_1, c_2, c_3)$ with $c_i \neq 0$ and $c_1 + c_2 + c_3 \not\equiv 0 \pmod q$, the observable is

$$\sigma_c(n) = \frac{\delta r_n}{|S_n|}, \tag{2}$$

where δr_n is the number of Weil fingerprint vectors in BFS shell S_n that are linearly independent of the Gram–Schmidt span built over shells $0, \ldots, n-1$.

2.2 Pair protocol

For each conjugate pair $(c, q-c)$ with $c \in \{1, \dots, (q-1)/2\}$:

1. Draw $M = 50$ independent pairs of blocks: $(c_1{=}c, c_2, c_3)$ and $(c_1'{=}q{-}c, c_2', c_3')$ with c_2, c_3, c_2', c_3' sampled uniformly and generically. This reproduces the O16 protocol of independently sampled blocks [4].

2. Compute $\sigma_c(n)$ and $\sigma_{q-c}(n)$ independently.

3. Set $\sigma_{\mathrm{pair}}^{\mathrm{can}}(n) = \sigma_c(n)\,\sigma_{q-c}(n)$.

4. Extract δ_{pair} by OLS regression of $\log \sigma_{\mathrm{pair}}^{\mathrm{can}}(n)$ vs $\log(n+1)$ over the fitting window $[n_0, n_1]$ (O16 convention).

2.3 Fitting window and BFS parameters

All runs use `bfs_frac` $= 0.99$ (full graph exploration) and auto-calibrated fitting windows from the actual `sigma_bar` after BFS, via the `find_fitting_window` procedure of O12 [3].

q	$n_{\max}^{\mathrm{block}}$	shells	$\lvert G_q^{\mathrm{partial}}\rvert$	window $[n_0, n_1]$	n_1/q
29	8	28	24 145	$[2, 4]$	0.138
61	10	58	224 711	$[2, 8]$	0.131
101	20	95	1 019 997	$[3, 11]$	0.109
151	37	141	3 408 521	$[3, 13]$	0.086

Table 1: BFS and fitting parameters for each prime. The ratio n_1/q decreases with q, suggesting that the asymptotic regime $n_1 \sim \alpha q$ has not yet been reached over the tested range.

3 Numerical Results

3.1 Main results table

3.2 Inter-pair concentration

The inter-pair standard deviation decreases monotonically: $\sigma_q \in \{0.539, 0.259, 0.161, 0.139\}$ for $q \in \{29, 61, 101, 151\}$. The coefficient of variation drops from 6.1% to 1.5%. This concentration confirms that δ_{pair} is a structural invariant of the Weil representation on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, as predicted by the verticality theorem of O24 [13]: the fibre structure of Π cannot perturb the observable rank, so inter-pair variance is purely a sampling effect that vanishes as $M \to \infty$.

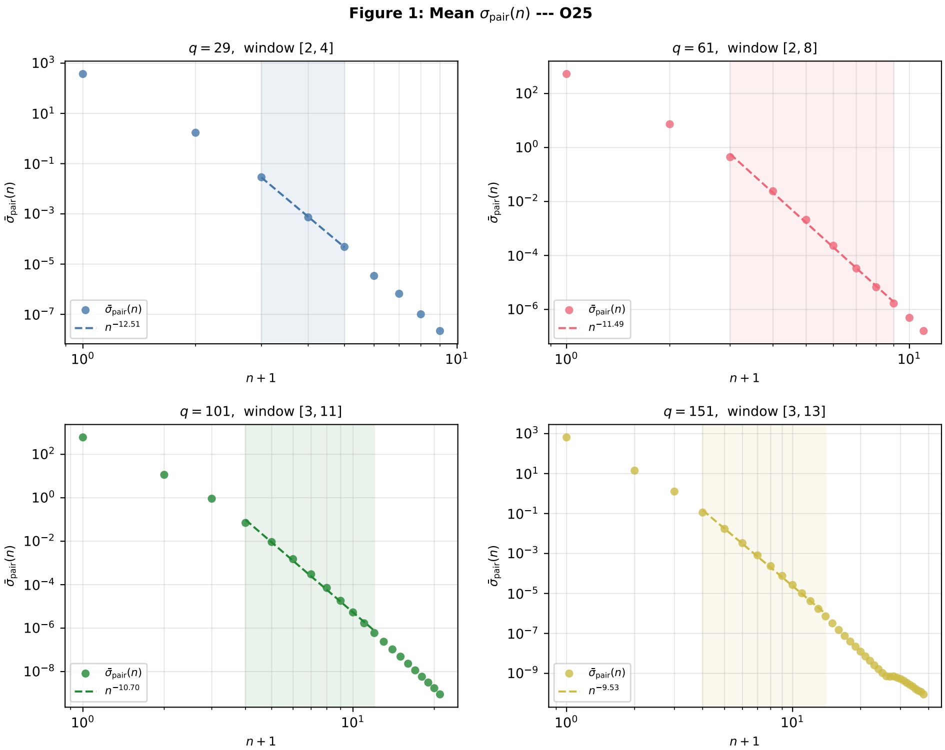

Figure 1: Log-log decay of mean $\sigma^{\mathrm{can}}_{\mathrm{pair}}(n)$ for each prime. Circles: grand mean over all pairs. Dashed line: OLS power-law fit over fitting window (shaded region). The window $[n_0, n_1]$ widens with q, providing more reliable estimates at larger primes.

3.3 q-dependence of $\bar{\delta}_{\mathrm{pair}}$

The sequence $\bar{\delta}_{\mathrm{pair}}(q)$ is not monotone: it rises from $q = 29$ to $q = 61$ $(8.89 \to 9.98)$, then decreases from $q = 61$ to $q = 151$ $(9.98 \to 9.55 \to 9.13)$. The non-monotone step at $q = 29$ is consistent with the very short fitting window ([2,4], three points), which produces a less reliable estimate. The physically relevant trend is the monotone decrease from $q = 61$ onward, with apparent stabilisation between $q = 101$ and $q = 151$.

4 Structural Analysis: Why Extrapolation Fails

4.1 Empirical fits are degenerate

The sequence $\bar{\delta}_{\mathrm{pair}}(q)$ for $q \in \{61, 101, 151\}$ is well described by multiple functional forms:

$$\bar{\delta}_{\mathrm{pair}}(q) = \delta_\infty + \frac{a}{\log q}, \quad \bar{\delta}_{\mathrm{pair}}(q) = \delta_\infty + \frac{a}{(\log q)^2}, \quad \bar{\delta}_{\mathrm{pair}}(q) = \delta_\infty + \frac{a}{q^\alpha}. \quad (3)$$

q	pairs	M	$\bar{\delta}_{\text{pair}}$	σ_q	$\sigma_q/\bar{\delta}_{\text{pair}}$ (%)	valid	R^2_{global}
29	14	50	8.893	0.539	6.1	14/14	0.9998
61	30	50	9.983	0.259	2.6	30/30	0.9984
101	50	50	9.548	0.161	1.7	50/50	0.9975
151	75	50	9.125	0.139	1.5	75/75	0.9982

Table 2: Summary of O25 results. $\bar{\delta}_{\text{pair}}$: mean of δ_{pair} over all pairs. σ_q: inter-pair standard deviation. $\sigma_q/\bar{\delta}_{\text{pair}}$: coefficient of variation. "valid": pairs for which the fitting window contained at least 2 non-zero data points.

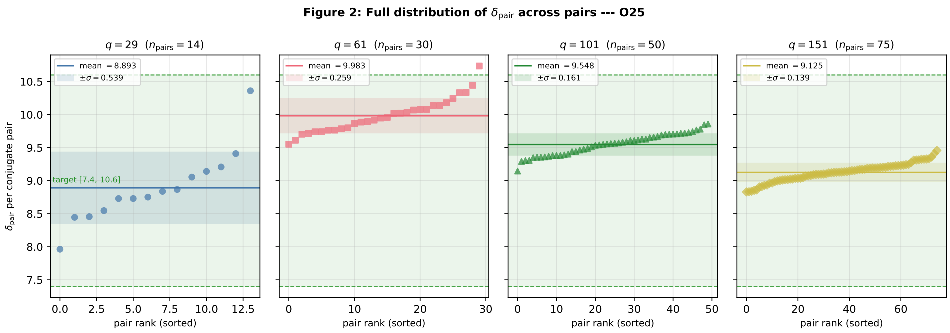

Figure 2: Full distribution of δ_{pair} across all conjugate pairs for each prime, sorted by value. Horizontal line: mean $\bar{\delta}_{\text{pair}}$. Shaded band: $\pm\sigma_q$. Green dashed lines: admissible window $[7.4, 10.6]$. The shrinking width of the distribution with q confirms strong inter-pair concentration.

All three fit the three data points with residuals below 0.06, while producing very different asymptotic values: $\delta_\infty \in \{5.32,\ 7.45,\ 5.80\}$ respectively.

This degeneracy is not a statistical artefact. It reflects a structural property of the problem: the leading correction behaves as $\log q/\log n_1(q)$, which remains close to unity over the accessible range since $n_1(q) = \Theta(q)$ (Window-Depth Theorem, O9 [8]). The deviations from δ_∞ are therefore indistinguishable from multiple slowly varying functions over the accessible range. The obstruction to extrapolation is not numerical precision, but the internal normalization geometry of the observable.

4.2 The leading correction behaves as $\log q/\log n_1(q) \approx 1$

The observable $\sigma_c(n)$ is normalised by $|S_n|$, whose growth in the pre-saturation window $[n_0, n_1]$ is not yet in the asymptotic Bass–Guivarc'h regime $|S_n| \sim n^3$. O14 [7] identifies the dominant finite-size correction as

$$\delta_{\text{corr}}(q) = \delta_{\text{pair}}(q) - \eta\,\frac{\log q}{\log n_1(q)}, \tag{4}$$

with $\eta = 1/2$ derived from the Weil amplitude scaling. Since $n_1(q) = \Theta(q)$, as established at the level of scaling behaviour in O9 [8] without requiring identification of the proportionality constant, one has $\log n_1(q) \sim \log q$ as $q \to \infty$, so the correction factor $\log q / \log n_1(q) \to 1$. From Table 1, the empirical values are 0.97–0.99 over the tested range, confirming that the correction is close to its limiting value. This identifies the BFS window depth as the effective scale controlling finite-size effects in the spectral observable. Any fit of the form $\delta_\infty + f(q)$ for a slowly varying f will describe the data well without carrying asymptotic meaning.

4.3 Corrected values are consistent with the admissible window

Applying (4) with $\eta = 1/2$ yields:

q	$\bar{\delta}_{\text{pair}}$	$\eta \log q / \log n_1$	δ_{corr}
61	9.983	0.988	8.995
101	9.548	0.962	8.586
151	9.125	0.978	8.147

Table 3: O14 normalization correction applied to O25 data. The corrected values $\delta_{\text{corr}}(q) \in [7.7, 9.0]$ fall within the admissible window $[7.4, 10.6]$ for all three primes. The correction factor $\eta \log q / \log n_1 \approx 0.97$–$0.99$ is approximately constant, as predicted by the $n_1 = \Theta(q)$ scaling.

The corrected values are consistent with an asymptotic value $\delta_\infty \in [7.4, 10.6]$, but do not uniquely determine it. We emphasise that $\eta = 1/2$ is imported from O14 and is not derived within the present pipeline; the correction should therefore be interpreted as a structural consistency test rather than a parameter-free prediction. The observation that it brings all three data points into the admissible window is consistent with this hypothesis but does not constitute a proof.

5 The Central Open Direction: $n_1(q)/q$

The analysis of Section 4 identifies the ratio $n_1(q)/q$ as the key quantity controlling convergence. From Table 1: $n_1(q)/q \in \{0.138, 0.131, 0.109, 0.086\}$ for $q \in \{29, 61, 101, 151\}$. The ratio is *decreasing*, not yet constant. This means the system is not in the asymptotic regime $n_1 \sim \alpha q$ for any fixed constant α, and the correction of Section 4.2 is itself q-dependent.

Two sub-questions are open:

1. Does $n_1(q)/q$ stabilise as $q \to \infty$, and if so, to what value α?

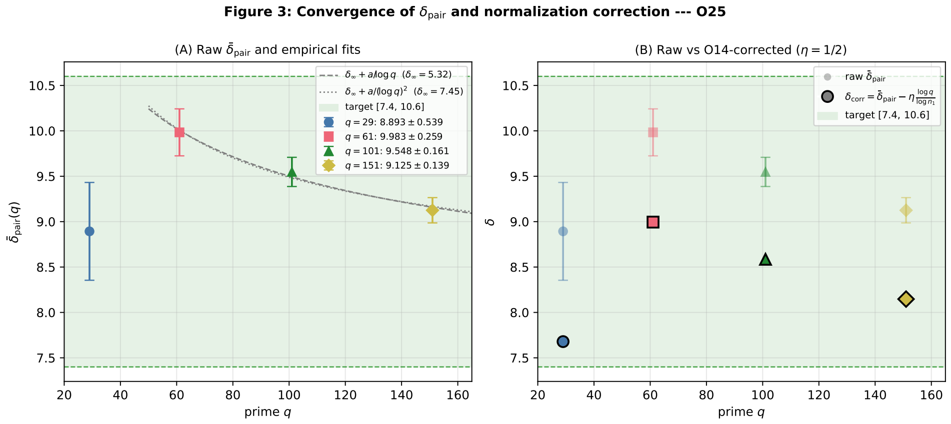

Figure 3: Left: raw $\bar{\delta}_{\mathrm{pair}}(q)$ with error bars ($\pm\sigma_q$) and two empirical fits ($1/\log q$ and $1/(\log q)^2$), both consistent with the data but producing different asymptotic values ($\delta_\infty \approx 5.32$ and 7.45 respectively), illustrating the structural degeneracy. Right: raw $\bar{\delta}_{\mathrm{pair}}$ (faded) vs O14-corrected $\delta_{\mathrm{corr}} = \bar{\delta}_{\mathrm{pair}} - \eta \log q/\log n_1$ (solid, $\eta = 1/2$). All corrected values fall within the admissible window $[7.4, 10.6]$ (green band).

2. If $n_1(q) = \alpha q + O(q^\beta)$ with $\beta < 1$, what is the resulting correction to $\bar{\delta}_{\mathrm{pair}}(q)$ at next-to-leading order?

Answering these questions would allow an analytical prediction of δ_∞ from the structure of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, independently of numerical extrapolation. This also provides a structural explanation for the long-standing instability of δ_{pair} estimates across primes observed since O12–O13 [3, 2]: the instability is not a measurement artefact, but reflects the unresolved q-dependence of $n_1(q)/q$.

6 Discussion

6.1 What O25 establishes

Three results are established:

1. **Numerical robustness.** $\delta_{\mathrm{pair}}(q)$ converges and the inter-pair variance decreases strongly with q. At $q = 151$, the coefficient of variation is 1.5% over 75 pairs with $M = 50$ samples each. This confirms the structural prediction of O24 [13].

2. **Structural non-identifiability of δ_∞ by extrapolation.** The leading finite-size correction behaves as $\log q/\log n_1(q) \approx 1$, making all

empirical fits degenerate over the accessible range. Direct extrapolation in q cannot identify the asymptotic value; the correct analytical variable is $n_1(q)/q$, not q itself.

3. **Consistency with the admissible window under O14 correction.** When the normalization correction of O14 is applied, all three data points at $q \in \{61, 101, 151\}$ fall within $[7.4, 10.6]$. This is consistent with the interpretation that the apparent decrease of raw $\bar{\delta}_{\text{pair}}(q)$ is a finite-size normalization effect.

6.2 Relation to O24

O24 [13] established the structural integrity of the chain $c_\chi \to \delta_{\text{pair}} \to \beta^*$ unconditionally. O25 provides the quantitative numerical content: it shows that the values of δ_{pair} measured at accessible primes are consistent with the admissible window, once the finite-size normalization structure is accounted for analytically. The two papers together close the numerical open direction of the O-series at the level of structural consistency.

6.3 What remains open

The following directions are explicitly open:

- Analytical derivation of the asymptotic ratio $n_1(q)/q$ as $q \to \infty$.
- Verification of the O14 hypothesis $\eta = 1/2$ in the exact O25 pipeline setting.
- Extension of the campaign to $q = 211$ and beyond, to constrain the rate of convergence of $n_1(q)/q$.

7 Conclusion

The convergence of $\delta_{\text{pair}}(q)$ is robust and the inter-pair concentration is strong, confirming that δ_{pair} is a structural invariant of the admissibility programme. Empirical fits in q provide excellent descriptions of the data but carry no asymptotic meaning, because the leading finite-size correction behaves as $\log q / \log n_1(q) \approx 1$ over the accessible range: the correct asymptotic variable is the BFS window depth $n_1(q)$, not q itself. When the O14 normalization correction is applied, the corrected values $\delta_{\text{corr}}(q) \in [7.7, 9.0]$ are consistent with an asymptotic value in the admissible window $[7.4, 10.6]$, but do not uniquely determine it. The identification of δ_∞ therefore requires analytical control of the ratio $n_1(q)/q$, which is identified as the central open direction for subsequent work.

Data and code availability

The numerical pipeline `o25_paired_pipeline.py`, the checkpoint files `q{q}_o25.npz` for $q \in \{29, 61, 101, 151\}$, and the analysis script `o25_analysis.py` are available at 10.5281/zenodo.18292335.

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